Information Aggregation and Market Feedback: Evidence from the introduction of NASDAQ's closing auction

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Abstract:
We examine how improvements in high-frequency price formation influence the feedback that financial markets provide to corporate decision makers, leveraging NASDAQ's introduction of a closing call auction designed to improve the accuracy of daily closing prices by more efficiently aggregating order flow and the information it contains. Although this change primarily affects intraday trading mechanics and the timing of information incorporation, it generates meaningful real effects. After the auction's adoption, corporate investment becomes more responsive to stock prices, especially for firms whose fundamentals are closely tied to macroeconomic conditions and growth opportunities, settings in which markets hold comparative informational advantages over managers. Additional analyses indicate greater managerial learning from prices: firms exhibit larger future earnings response coefficients, management forecasts become more sensitive to contemporaneous returns, and operating performance improves. Overall, the evidence suggests that improved information aggregation, beyond information processing or production, strengthens market feedback and shapes real economic decisions.

JEL Classifications: G14, G31, M40, M41
Keywords: Information aggregation, Closing auction, Market feedback, Managerial learning, Investment-*q* sensitivity, Future earnings response coefficient, Management forecasts

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1. Introduction

Secondary financial markets perform two main functions: providing liquidity and facilitating price discovery. Price discovery is the process by which new information about fundamental value becomes reflected in asset prices (O'Hara, 2003). This dynamic and complex process involves diverse participants who process, produce, and trade on information that is ultimately aggregated into prices.[1] A growing literature shows that price discovery is not merely a sideshow but actively generates new information that guides real economic decisions (e.g., Luo, 2005; Jayaraman and Wu, 2019), a phenomenon known as *market feedback* (Goldstein, 2023).[2] Prior studies document how institutional factors shape market feedback by influencing how informed investors process and produce information.[3] Yet, despite being indispensable to price discovery, the role of *information aggregation* during price formation in driving market feedback remains largely unexplored. We address this gap by exploiting NASDAQ's reform of its closing price formation mechanism.

In April 2004, NASDAQ introduced a call auction at market close (hereafter, the *closing auction*) alongside its continuous dealer market to set closing prices. Previously, NASDAQ

[1] Blankespoor, deHaan, and Marinovic (2020) define information processing as the process of monitoring, acquiring, and analyzing information.
[2] It is also widely referred to as the feedback effect or managerial learning (Goldstein, 2023). In this paper, we use these terms interchangeably.
[3] Factors that influence market feedback by affecting informed investors' incentives or costs to process or produce information include cross-listing (Foucault and Fresard, 2012), insider trading enforcement (Edmans, Jayaraman, and Schneemeier, 2017), mandatory corporate disclosure (Jayaraman and Wu, 2019), dissemination of corporate filings via SEC EDGAR (Bird, Karolyi, Ruchti, and Truong, 2021), tick size changes (Ye, Zheng, and Zhu, 2023), and mandatory portfolio disclosure (Sani, Shroff, and White, 2023).

closing prices were based on the last trade of the continuous session. Under the new system, buy and sell orders, so-called *on-close orders*, are accumulated and executed simultaneously at a single market-clearing price that maximizes trading volume at the close. NASDAQ also began disseminating information about on-close order imbalances during the last ten minutes of trading, allowing participants to offset imbalances with opposite orders.

The NASDAQ closing auction was introduced to enhance end-of-day price formation, with the goal of ensuring that daily closing prices more effectively aggregate market information (Smith, 2006; Pagano, Peng, and Schwartz, 2013). Consistent with this objective, prior market microstructure research shows that the closing auction improves high-frequency price efficiency, typically reflected in reduced noise in intraday or daily prices (e.g., Pagano et al., 2013). Although more accurate stock prices are generally expected to strengthen managerial learning, it remains an open question whether improvements in high-frequency price efficiency actually translate into better learning and ultimately lead to more efficient real investment decisions. This uncertainty arises because market design features can influence the accuracy of closing prices and their informativeness for managerial learning in different, and sometimes conflicting, ways.[4]

Indeed, indirect evidence suggests that the closing auction may actually *reduce* the amount of firm-specific information incorporated into prices. Pagano and Schwartz (2005) document that return non-synchronicity, commonly interpreted as a proxy for the amount of private information embedded in stock prices (e.g., Morck, Yeung, and Yu, 2000; Chen, Goldstein,

[4] For example, Weller (2018) finds that although algorithmic trading helps incorporate available information into prices, it reduces overall price informativeness. Similarly, Ye et al. (2023) show that while a larger tick size generally increases price noise, it nonetheless enhances managerial learning.

and Jiang, 2007), declined following NASDAQ's adoption of the closing auction.[5] A similar decline was observed after the Paris Bourse introduced its own closing auction (Pagano and Schwartz, 2003). Taken together, these patterns raise the possibility that, even though the auction improves closing-price accuracy, it may simultaneously weaken the informativeness of the price signals that managers rely on to guide real decisions.

Theory also offers mixed predictions. The closing auction primarily alters *how* and *when* information is aggregated into prices, but it does not necessarily affect the total *amount* of information ultimately produced by traders. When the production of private information in the market is relatively limited, shifting the timing of order matching—as the closing auction does—may have little impact on the overall information incorporated into prices over longer horizons. In this case, the auction would not meaningfully strengthen or weaken managers' ability to learn from stock prices. However, when information production is more continuous, market design can influence price informativeness through two opposing channels. On one hand, consolidating orders into a multilateral closing auction reduces adverse-selection risk and enables informed traders to hide within a larger pool of orders (Madhavan, 1992; Foucault, Kadan, and Kandel, 2005; Roşu, 2009), thereby encouraging informed trading and enhancing the incorporation of private information into prices. On the other hand, the NASDAQ auction increases transparency by publicly disseminating on-close order imbalances during continuous trading, which may discourage informed traders from participating (Biais, Hillion, and Spatt, 1999) and potentially slow the flow of private information into prices.

Given these competing theoretical channels and the mixed empirical evidence, it

[5] Return non-synchronicity is defined as one minus adjusted R^2 from regressions of individual stock returns on market returns.

ultimately remains an empirical question whether NASDAQ's introduction of the closing auction strengthened or weakened the price signals available to managers for making investment decisions. To examine the effect of NASDAQ's closing auction on market feedback, we employ a difference-in-differences (DiD) design that leverages its staggered phase-in between April 12 and December 13, 2004. The auction's mandatory implementation helps mitigate endogeneity concerns arising from unobserved firm-, industry-, or market-level shocks, as well as from reverse causality (Goldstein, Yang, and Zuo, 2023; Leuz and Wysocki, 2016). For an omitted variable to confound our results, it would have to systematically affect different firm cohorts precisely according to the phase-in schedule.

Because market feedback is not directly observable, we follow prior research (e.g., Chen et al., 2007; Edmans et al., 2017; Jayaraman and Wu, 2019) and infer it from the sensitivity of future investment to current q, commonly referred to as investment-q sensitivity. The intuition behind this measure is that investment-q sensitivity should be stronger when stock price movements reflect information that is *new* to managers rather than information they already possess (Chen et al., 2007). Accordingly, if the closing auction causes prices to aggregate more (less) private information new to managers, investment-q should increase (decrease).

We find that investment-q sensitivity rises significantly, by about 7.84%, after the implementation of the closing auction, relative to the pre-auction level. This effect remains robust after controlling for firm heterogeneity and group-specific time trends in investment-q sensitivity, financial constraints, and the contemporaneous introduction of mandatory portfolio disclosure (Sani et al., 2023). The result also holds under two alternative specifications designed to address potential bias from using already treated firms as controls in the staggered DiD

design: a stacked DiD regression (Cengiz, Dube, Lindner, and Zipperer, 2019) and a standard DiD regression using a matched control group of NYSE-listed firms with similar characteristics. Moreover, a dynamic test shows no significant pre-trend differences in investment-*q* sensitivity between treatment and control groups, consistent with the parallel-trends assumption. Together, these results support the inference that the observed increase in investment-*q* sensitivity is driven by the introduction of the closing auction.

Although a higher investment-*q* sensitivity is consistent with stronger market feedback and managerial learning, alternative explanations may also account for the pattern.[6] A key challenge in isolating market feedback lies in the absence of a direct proxy for managerial learning (Edmans et al., 2017). To reinforce the feedback interpretation, we examine the underlying mechanism, referred to as the *feedback channel*, through which the closing auction enhances market feedback. Specifically, the auction aggregates more private information into stock prices, some of which is new to managers. Consequently, managers rely more on prices as an information source for investment decisions, leading to higher investment-*q* sensitivity. A series of tests provides evidence consistent with this feedback channel.

We first examine the impact of the closing auction on the future earnings response coefficient (FERC), which captures the extent to which current stock returns anticipate future earnings (e.g., Israeli, Lee, and Sridharan, 2017; Lee and Watts, 2021). Brogaard and Pan (2022) show that FERC rises with the amount of private information about future earnings contained in prices. We find that FERC increases after the closing auction's implementation, consistent with the interpretation that the auction enhances the aggregation of private information into

[6] In Section 6, we rule out three alternative explanations for our findings: increased information acquisition, reduced price noise, and decreased managerial information.

stock prices.

Next, we assess whether this enhanced information aggregation provides managers with new information. We first infer new information from changes in managerial beliefs, captured by revisions to management earnings forecasts. While publicly disclosed management earnings forecasts are still proxies for internal managerial expectations, the link between the two is likely to be strong (Hemmer and Labro, 2008; Dichev, Graham, Harvey, and Rajgopal, 2013; Zuo, 2016). Using Zuo's (2016) research design, we find that after the implementation of the closing auction, management forecast revisions become more sensitive to contemporaneous stock returns, suggesting that managers glean more new information from stock prices to update their beliefs.

We further infer new information based on its type. The feedback literature posits that information about external conditions, such as macroeconomic factors and growth opportunities, is more likely to be new to managers (Bond et al., 2012; Bai, Philippon, and Savov, 2016; Ye et al., 2023). Consistent with this view, the increase in investment-*q* sensitivity is more pronounced among firms whose earnings co-move more with macroeconomic factors (e.g., GDP and energy prices) and among firms that are more exposed to growth opportunities (i.e., growth firms), but is absent among value firms. These patterns suggest that the closing auction improves the aggregation of information about growth opportunities and macroeconomic conditions into prices.

Finally, we examine whether stronger market feedback translates into better future performance. If managers incorporate newly aggregated information into investment decisions, investment efficiency should improve, enhancing firm outcomes (e.g., Chen et al. 2007;

Jayaraman and Wu, 2019, 2020). Consistent with this prediction, we find significant post-auction increases in operating performance, measured by return on assets and operating profits.

Our paper contributes to two strands of literature. First, we add to the literature on the feedback effect of financial markets (see Bond et al., 2012; Goldstein, 2023). Prior studies have focused primarily on how the incentives and costs faced by informed traders in processing and producing information shape market feedback, largely overlooking the role of information aggregation. The effect of aggregation mechanisms on feedback is *ex ante* ambiguous, given the mixed theoretical and empirical evidence. Exploiting NASDAQ's introduction of the closing auction, we provide causal evidence that the mechanism of information aggregation in price formation plays a significant role in shaping market feedback. Our findings highlight the importance of considering information aggregation, beyond information processing and production, when studying market feedback and managerial learning.

Second, we contribute to the literature on the economic consequences of closing auctions and pricing mechanisms more broadly. Closing auctions have gained increasing importance as they account for a growing share of trading volume and determine reference prices crucial for corporate decisions and market activities.[7] Existing research has focused mainly on their effects on liquidity and short-horizon price efficiency. We extend this literature by examining their *real* effects, addressing Goldstein's (2023) call to evaluate institutional and technological innovations in financial markets through the lens of market feedback. We show that the

[7] Closing prices serve several critical market functions. They are used to convert mutual fund inflows and withdrawals into holdings and cash, calculate the net asset values of exchange-traded funds and stock indices, benchmark institutional investor performance, and settle derivative contracts (Bogousslavsky and Muravyev, 2023; Pagano et al., 2013). Closing prices on executive stock compensation grant dates are also used to determine fair value. Moreover, daily closing prices, widely disseminated by data vendors such as CRSP and Yahoo Finance, underpin numerous financial analyses and applications.

introduction of the closing auction enhances managerial learning from stock prices, resulting in more efficient investment decisions and stronger operating performance. Our results further suggest that the post-auction decline in return non-synchronicity documented by Pagano and Schwartz (2003, 2005) does not imply weaker information aggregation at the close.

The remainder of the paper is organized as follows: Section 2 describes the NASDAQ closing auction and reviews related literature. Section 3 outlines the research design, data, and sample selection. Section 4 presents results on the impact of the NASDAQ closing auction on investment-*q* sensitivity, followed by tests of the feedback channel in Section 5. Section 6 reports additional analyses, and Section 7 concludes.

2. Institutional Background and Literature Review

2.1. NASDAQ closing auction

In April 2004, NASDAQ introduced the *Closing Cross*, a call auction designed to determine official closing prices for all NASDAQ-listed securities. Previously, the closing price was simply determined by the last transaction of the trading day, regardless of size (Smith, 2006), raising concerns about representativeness and vulnerability to manipulation.[8] Major institutions, including S&P and index funds, pushed for a more reliable price discovery mechanism. In response, NASDAQ introduced the closing auction with the goal of producing more accurate and reliable closing prices that better reflect true market supply and demand.[9]

[8] Market participants, including fund managers, might have incentives to manipulate closing prices for various reasons. These include inflating quarter-end performance (Carhart, Kaniel, Musto, and Reed, 2002), profiting from positions in derivatives linked to the underlying stocks (Ni, Pearson, and Poteshman, 2005), and exploiting discrepancies between a funds' fundamental value and its net asset value (NAVs). Hillion and Suominen (2004) documented a significant increase in price volatility during the last minute of trading before the Paris Bourse introduced a closing call auction in 1998. Additionally, regulators prosecuted a total of 184 cases of price manipulation between January 1997 and January 2009 (Comerton-Forde and Putniņš, 2011).

[9] https://www.nasdaq.com/solutions/nasdaq-closing-cross.

Unlike continuous trading, where orders are matched sequentially, the closing auction accumulates all orders and executes them at one market-clearing price at 4:00 p.m. Traders may submit limit-on-close (LOC) and market-on-close (MOC) orders until 3:50 p.m.,[10] after which modifications are prohibited. To manage potential order imbalances, NASDAQ also introduced Imbalance Only (IO) orders, accepted from 3:30 p.m. until the auction,[11] which convert into limit orders at the best bid or ask at 4:00 p.m.[12] To enhance transparency, NASDAQ disseminates Order Imbalance Information (NOII) from 3:50 p.m. onward,[13] providing real-time data on order imbalances, reference prices, and paired shares to help participants assess market supply and demand.

At 4:00 p.m., all eligible orders are crossed at the price that maximizes executed volume. This price becomes the NASDAQ Official Closing Price (NOCP), reported on the consolidated tape. Figure 1 illustrates the auction mechanism. By aggregating and matching orders multilaterally, the closing auction is designed to produce a more accurate and representative closing price than the previous last-trade method. The auction was phased in beginning April 2004 with a small group of pilot stocks. By mid-June, all S&P 500 and NASDAQ 100 stocks were included, and by December 13, 2004, the mechanism covered all NASDAQ-listed stocks.

[10] As of October 2018, the deadline for submitting MOC (LOC) orders was changed to 3:55:00 p.m. (3:58:00 p.m.)
[11] Beginning in March 2013, IO, LOC, and MOC orders could be submitted as early as 4:00 a.m.
[12] Buy (sell) IO orders priced more aggressively than NASDAQ's best bid (ask) at 4:00:00 p.m. are re-priced to the best bid (ask) prior to execution in the closing cross.
[13] NASDAQ disseminates the NOII every 30 seconds until 3:55:00 p.m., every 15 seconds until 3:59:00 p.m., and every 5 seconds thereafter, with the final update released at 3:59:50 p.m. The start time for NOII dissemination was moved to 3:55:00 p.m. in October 2018.

Appendix A details the rollout timeline.

2.2. Literature review

2.2.1. Closing Auction

Market microstructure research has long explored how different price formation mechanisms, such as continuous quote-driven systems (e.g., dealer markets), continuous order-driven systems (e.g., limit order books), and periodic order-driven systems (e.g., call auctions), affect liquidity and price discovery.[14] Among these, closing auctions have drawn particular interest because closing prices are key reference points for portfolio valuation, index benchmarking, and derivative settlement. Prior studies have mainly examined their effects on short-horizon price efficiency, with only limited indirect evidence on whether they influence managerial learnings and market feedback.[15]

Extensive evidence shows that closing auctions reduce microstructure noise – temporary price deviations from fundamental values caused by trading frictions. Pagano et al. (2013) document a sharp decline in return volatility immediately before market close after NASDAQ introduced its closing auction, attributing it to reduced price noise. Similarly, Smith (2006) finds that NASDAQ's closing auction lowers price dispersion just before the close and weakens close-to-open reversals. Comparable results are reported for the Singapore Exchange and Borsa Italiana (Chang, Rhee, Stone, and Tang, 2008; Comerton-Forde, Lau, and McInish, 2007;

[14] In a quote-driven system, liquidity providers (typically market makers or dealers) post bid and ask prices in advance, while in an order-driven system, traders submit buy and sell orders to a central order book, and prices are determined by the interaction of these orders. In a continuous market, an investor's order is executed immediately upon submission. In contrast, a periodic system accumulates investors' orders for simultaneous execution at a pre-determined time.

[15] The short-horizon price efficiency measures commonly used in the literature generally align with forecasting price efficiency (FPE) as defined by Bond et al. (2012), which reflects the extent to which stock prices predict or explain firms' fundamental values. In contrast, managerial learning and market feedback relate to revelatory price efficiency (RPE), which captures the degree to which stock prices convey new information to decision makers.

Kandel, Rindi, and Bosetti, 2012).

Despite these improvements, inefficiencies at the close persist, though they are short-lived. Bogousslavsky and Muravyev (2023) show that small deviations between auction prices and 4:00 p.m. midpoint quotes typically reverse the next morning, while Jegadeesh and Wu (2022) find that returns from the first NOII dissemination to close reverse over three to five days. Both attribute these reversals to order imbalances from uninformed investors such as index funds and ETFs.[16]

Beyond reducing noise, an important dimension of price quality is the extent to which prices efficiently incorporate public information. Evidence on this dimension in the context of closing auctions remains limited and mixed. Pagano and Schwartz (2003) show that introducing closing auctions on the Paris Bourse reduced price delays, lags in individual stock returns' adjustment to market-wide information, for Continuous A stocks but not for Continuous B stocks. In contrast, Indriawan, Pascual, and Shkilko (2024) find no significant change in intraday price delays following the Taiwan Stock Exchange's 2020 transition from frequent batch auctions to a continuous limit order book.

To the best of our knowledge, no prior study directly examines the effect of closing auctions on information aggregation and managerial learning. The closest indirect evidence comes from Pagano and Schwartz (2003, 2005), who document a significant decline in return non-synchronicity following the introduction of closing auctions on NASDAQ and the Paris Bourse. If return non-synchronicity is interpreted as capturing the amount of private

[16] Jegadeesh and Wu (2022) and Bogousslavsky and Muravyev (2023) argue that noise in closing prices is likely driven by large order imbalances from uninformed index funds and exchange-traded funds (ETFs) at market close. However, Antoniou, Li, Liu, Subrahmanyam, and Sun (2023) challenge the assumption that all ETFs are uninformed. They show that non-market ETFs positively affect real investment efficiency by facilitating greater information flows, suggesting that these ETFs are informed and can provide valuable new information to managers.

information embedded in prices (e.g., Morck et al., 2000; Chen et al., 2007), this decline could suggest that closing auctions weaken private information aggregation and, in turn, managerial learning. However, return non-synchronicity also decreases when price noise falls, because lower noise mechanically increases return comovement with the market. Thus, these findings could imply that closing auctions reduce private information in prices, improve price accuracy by reducing noise, or both. Our study helps disentangle these competing interpretations by examining longer-horizon information content and by providing direct evidence on how closing auctions affect managerial learning.

2.2.2. Market Feedback

Research on market feedback rests on two key premises. First, market prices aggregate information from diverse participants (Hayek, 1945). Second, some of this information is new to decision-makers, making prices informative signals for guiding real decisions. Ample empirical evidence supports this view. Luo (2005) finds that managers are more likely to cancel acquisitions after negative market reactions. Zuo (2016) shows that stock prices convey information about firms' fundamentals unknown to managers, which they then incorporate into earnings forecasts. Similarly, Jayaraman and Wu (2020) document that managers adjust capital expenditures in response to market reactions to their initial capex forecasts.

Recent studies explore how institutional and technological factors shape market feedback by influencing informed investors' information production and processing, thereby affecting the amount of new information managers can glean from stock prices. Factors that strengthen feedback by encouraging private information production include stricter insider trading enforcement (Edmans et al., 2017), reduced tick size (Ye et al., 2023), and inclusion in

exchange-traded funds (Antoniou et al., 2023). Conversely, factors that weaken feedback by crowding out private information production include mandatory segment disclosure (Jayaraman and Wu, 2019), portfolio disclosure by actively managed investment funds (Sani et al., 2023), and the electronic dissemination of corporate filings through the SEC's EDGAR system (Bird et al., 2021).

Our study extends this literature by demonstrating that the feedback effect also depends on information aggregation, an indispensable component of price discovery. Showing that multiple components of the price discovery process influence managerial learning provides triangulating evidence of market feedback.

3. Research design, data, and sample selection

To examine the effect of the NASDAQ closing auction on market feedback, we estimate a staggered DiD regression, as specified in equation (1). The model extends the investment-*q* regression by including an interaction term that captures the staggered adoption of the closing auction:

$$Inv_{i,t+1} = \beta_1 Q_{i,t} \times Auction_{i,t} + \beta_2 Q_{i,t} + \beta_3 Auction_{i,t} + \gamma X_{i,t} + \alpha_i + \eta_t + \varepsilon_{i,t+1} \quad (1)$$

where $Inv_{i,t+1}$ represents firm *i*'s investment in quarter *t*+1, defined as the sum of capital expenditures and R&D expenses scaled by beginning-of-quarter book assets (e.g., Chen et al., 2007; Antoniou et al., 2023).[17] $Q_{i,t}$ represents firm *i*'s Tobin's *q*, measured at the end of quarter *t*. It is calculated as the market value of equity plus the book value of assets minus the book value of equity, scaled by book assets. We calculate market value of equity using closing prices

[17] see Appendix B for variable definitions.

from the Trade and Quote (TAQ) database multiplied by common shares outstanding.[18] $Auction_{i,t}$ is an indicator equal to one if a firm has been phased into the NASDAQ closing auction by the end of quarter t, and zero otherwise. Our coefficient of interest, β_1 on $Q_{i,t}\times Auction_{i,t}$, captures the change in investment-q sensitivity following the closing auction's implementation.

Turning to control variables ($X_{i,t}$) in equation (1), we include cash flows and its interaction with the auction variable ($CF_{i,t}$ and $CF_{i,t}\times Auction_{i,t}$), given the documented effects of cash flows on investment (e.g., Fazzari, Hubbard, and Petersen, 1988). As discussed in Section 2.1, the phase-in of the NASDAQ closing auction was not random. It began with index constituents and later expanded to smaller, less liquid firms with lower institutional ownership. Although non-random treatment assignment is common in regulatory settings and does not necessarily undermine causal inference (e.g., Gao and Huang 2020; Goldstein et al., 2023), we control for factors likely correlated with both treatment timing and investment-q sensitivity (e.g., Rösch, Subrahmanyam, and van Dijk, 2022). Specifically, we include firm size ($Size_{i,t}$), institutional ownership ($InstHold_{i,t}$), and Amihud's (2002) illiquidity measure ($ILLIQ_{i,t}$), along with their interactions with Tobin's q ($Q_{i,t}$) (deHaan, Moon, Shipman, Swanquist, and Whited, 2023). Finally, we include firm fixed effect (α_i) to control for time-invariant heterogeneity across firms and year-quarter fixed effect (η_t, hereafter referred to as quarter fixed effects) to absorb time-varying shocks. All unbounded variables are winsorized at the 1st and 99th percentiles to mitigate the influence of outliers. Standard errors are clustered at the firm level, and statistical significance is based on two-tailed tests.

[18] Section OA.1 of the Online Appendix details the procedure for extracting closing prices from TAQ.

We construct the variables for equation (1), using data from multiple sources: Compustat for financials, CRSP for daily and monthly stock data, TAQ for closing and intraday prices, I/B/E/S for management and analyst forecasts, and Thomson Reuters for insider trading and institutional ownership.

To construct our sample, we begin with NASDAQ-listed common stocks in the intersection of Compustat and CRSP. Following prior studies on investment-*q* sensitivity (e.g., Chen et al., 2007; Jayaraman and Wu, 2019), we exclude firms in the financial (SIC codes 6000-6999) and utility (SIC codes 4200-4299) industries. We also drop firms that changed their listing exchange during our sample period, as the NASDAQ closing auction applies only to NASDAQ-listed firms. Following Chen et al. (2007), we exclude firms with book equity below $10 million or a closing price below $1 at the end of 2003, the last year before the auction rollout. For the remaining firms, we retain quarterly data from 2002Q1 (two years before the first auction phase-in) through 2006Q4 (two years after the final phase-in). We require non-missing data for all variables needed to estimate investment-*q* sensitivity. The final sample consists of 28,521 firm-quarter observations for 1,644 NASDAQ-listed firms.

Table 1 reports summary statistics for the variables in our main sample. The average quarterly total investment (*CAPXRND*) equals 3.29% of beginning-of-quarter total assets, or 13.16% on an annualized basis. The mean value of *Auction* is 0.451, indicating that approximately 45% of firm-quarters are after the implementation of the closing auction. This proportion is close to 0.50 by design, as the sample includes roughly equal observations before and after the auction's introduction. The mean values of Tobin's *q* (*Q*) and cash flows (*CF*) are 2.209 and 0.234%, respectively. We discuss the remaining variables in Table 1 in subsequent

sections, where they are formally defined.

4. The Effect of the NASDAQ closing auction on investment-*q* sensitivity

4.1. Baseline regression

In this section, we estimate the treatment effects of the NASDAQ closing auction on investment-*q* sensitivity. Column (1) of Panel A in Table 2 presents results from estimating equation (1). The coefficient on $Q_{i,t}\times Auction_{i,t}$ is 0.130 and statistically significant, indicating a 7.84% (=0.13/1.659) increase relative to the pre-auction investment-*q* sensitivity after controlling for the effects of firm size and other factors. This finding suggests that the implementation of the closing auction leads to a substantial, yet economically plausible, increase in investment-*q* sensitivity among NASDAQ-listed firms. In columns (2) and (3) of Panel A, we decompose total investment (*CAPXRND*) into capital expenditures (*CAPX*) and R&D expenses (*RND*). Both regressions yield significantly positive coefficients on $Q_{i,t}\times Auction_{i,t}$, suggesting that both capital expenditures and R&D expenses become more responsive to Tobin's *q* after the implementation of the closing auctions.

The identification of the DiD estimation in equation (1) relies on the assumption that, absent treatment, the dependent variable would have followed parallel trends across treatment and control firms. Although this assumption cannot be tested directly, we assess its plausibility by examining pre-treatment dynamics in investment-*q* sensitivity. Following Foucault and Frésard (2012) and Goldstein et al. (2023), we replace $Auction_{i,t}$ in equation (1) with a series of event-time dummies defined relative to the quarter of each firm's auction rollout, using the quarter immediately preceding the rollout as the reference period.

Panel A of Figure 2 plots the coefficients on the interactions between event-time

dummies and Tobin's q (dots), along with 90% confidence intervals (dashed lines), for event times ranging from four or more quarters before ($k = -4$) to four or more quarters after ($k = 4$) the auction implementation ($k = 0$). Consistent with the parallel-trends assumption, investment-q sensitivity for treatment and control firms tracks each other closely prior to the auction rollout, with differences never statistically distinguishable from zero. Beginning in the first post-treatment quarter, however, the difference becomes significantly positive. Panels B and C show similar patterns for *CAPX* and *RND* sensitivities, respectively. Overall, Figure 2 implies that any omitted factor would need to coincide precisely with the auction's implementation to account for these results.

Turning to the control variables in column (1) of Panel A, Table 2, the coefficient on $CF_{i,t}$ is significantly negative. Although this contrasts with the conventional expectation of a positive investment-cash flow relationship, it aligns with Antoniou et al. (2023), who report similar evidence. One possible interpretation is that higher cash flows may signal lower growth opportunities for NASDAQ-listed firms during our sample period. Notably, the significantly positive coefficient on $CF_{i,t} \times Auction_{i,t}$ offsets the negative main effect, suggesting that investment becomes less sensitive to cash flows following the introduction of the closing auction.

Regarding factors potentially influencing firm assignment to different phase-in groups, neither firm size (*Size*) nor institutional ownership (*InstHold*) significantly affects investment-q sensitivity. Consistent with Rösch et al. (2022) and Amihud and Levi (2023), investment-q sensitivity increases with illiquidity (*ILLIQ*), as evidenced by the significantly positive coefficient on $Q_{i,t} \times ILLIQ_{i,t}$. Thus, the effect of the closing auction on investment-q sensitivity appears both distinct from and incremental to that of illiquidity.

While *Size*, *InstHold*, and *ILLIQ* likely capture most of the non-randomness in the timing of closing auction implementation among NASDAQ firms, we conduct additional robustness tests in Panel B to further address potential omitted correlated variables. In column (1), we include interactions between Tobin's *q* and firm fixed effects ($Q_{i,t}$×Firm FE) as well as between Tobin's *q* and group-specific time trends ($Q_{i,t}$×Group FE×Time trend), where groups comprise firms sharing the same auction implementation month. The coefficient on $Q_{i,t}$×$Auction_{i,t}$ remains statistically and economically significant, indicating that our main findings are not driven by unobserved firm heterogeneity or group-specific time trends in investment-*q* sensitivity.

In column (2), we control for financial constraints using the four-variable Kaplan-Zingales index, $KZ4_{i,t}$ (Baker, Stein, and Wurgler, 2003; Chen et al., 2007), and its interaction with Tobin's *q*. The coefficient on $Q_{i,t}$×$Auction_{i,t}$ remains largely unchanged, ruling out the possibility that financial constraints, rather than the closing auction, explain the observed change in investment-*q* sensitivity.

Finally, in column (3), we account for the potential confounding effect of the May 2004 SEC regulation mandating more frequent portfolio disclosures by actively managed funds. Sani et al. (2023) document that this regulation reduces investment-*q* sensitivity for firms with substantial ownership by affected funds. Because the regulatory change overlaps with NASDAQ's auction rollout, we follow their approach to identify treated firms, those with above-median ownership by actively managed funds in the quarter preceding the regulation, and define two indicator variables: MPD_Treat_i and MPD_Post_t, equal to one for treated firms and post-regulation quarters, respectively. We include interaction terms $Q_{i,t}$×MPD_Treat_i×MPD_Post_t,

$Q_{i,t}$×MPD_Treat_i, $Q_{i,t}$×MPD_Post_t, and MPD_Treat_i×MPD_Post_t as additional controls. As shown in column (3), the positive coefficient on $Q_{i,t}$×$Auction_{i,t}$ becomes even stronger, suggesting that the closing auction effect on investment-*q* sensitivity is distinct from the impact of mandatory portfolio disclosure.[19]

4.2. Alternative specifications

Recent studies highlight that staggered DiD regressions, such as equation (1), can yield biased estimates when treatment effects are delayed or heterogeneous, as early-treated firms implicitly serve as controls for later-treated firms (e.g., Baker, Larcker, and Wang, 2022; Barrios, 2022). To mitigate this concern, we estimate two alternative specifications that rely on "clean" control groups.

First, we implement a stacked DiD regression in which the control group is limited to the set of to-be-treated firms (e.g., Gormley and Matsa, 2011; Gormley, Matsa, and Milbourn, 2013; Cengiz et al., 2019). Specifically, for each phase-in group between April and June 2004, we form a separate cohort of treatment and control firms. For example, for the May 2004 phase-in group, we construct a cohort centered on May 2004, where treatment firms are those phased in during May 2004 and control firms are those scheduled for phase-in in November and December 2004 (i.e., to-be-treated firms). For both treatment and control firms, we collect data for the event window [−1, +1], where quarter 0 corresponds to the fiscal quarter including May 2004.[20] We then stack all cohorts and estimate the pooled regression. Because a control firm may appear in multiple cohorts, we follow prior studies by including cohort-firm and cohort-

[19] The coefficient on $Q_{i,t}$×MPD_Treat_i×MPD_Post_t in column (3), while negative, is not statistically significant. However, when the dependent variable is capital expenditures (rather than total investment), the coefficient becomes significantly negative, consistent with Sani et al. (2023).

[20] We drop a control firm from a cohort if its tracking quarter +1 includes the month in which the firm was phased into the closing auction.

quarter fixed effects and clustering standard errors at the cohort-firm level. The results, reported in column (1) of Table 3, continue to indicate a positive effect of the closing auction on investment-q sensitivity, as evidenced by the significantly positive coefficient on $Q_{i,t}\times Treat_i\times Post_{i,t}$.

Second, we employ a standard DiD estimator using propensity-scored-matched (PSM) NYSE-listed firms as the control group. These firms experience no change in their closing price determination during the sample period. Specifically, we match each NASDAQ-listed firm to a NYSE-listed firm within the same Fama-French 48 industry based on market capitalization, Tobin's q, and index fund ownership, using a nearest-neighbour propensity score method. Firm characteristics are measured at the end of 2003Q4 to avoid potential effects of the closing auction. NASDAQ-listed firms are classified as the treatment group ($Treat_i = 1$) and their matched NYSE-listed counterparts as the control group ($Treat_i = 0$).[21] Using the matched sample of 364 NASDAQ-NYSE firm pairs, we estimate the following standard DiD regression:

$$\begin{aligned} Inv_{i,t+1} = {} & \beta_1 Q_{i,t}\times Treat_i\times Post_{i,t} + \beta_2 Q_{i,t} + \beta_3 Q_{i,t}\times Treat_i + \beta_4 Q_{i,t}\times Post_{i,t} \\ & + \beta_5 Treat_i\times Post_{i,t} + \gamma X_{i,t} + \alpha_i + \eta_t + \varepsilon_{i,t+1} \end{aligned} \tag{2}$$

where $Post_{i,t}$ equals one for a matched pair once the NASDAQ treatment firm has been phased into the closing auction by the end of quarter t, and zero otherwise. The coefficient β_1 on $Q_{i,t}\times Treat_i\times Post_{i,t}$ captures the effect of the NASDAQ closing auction on investment-q sensitivity. The control variables ($X_{i,t}$) are identical to those in equation (1), except that we replace $CF_{i,t}\times Auction_{i,t}$ with $CF_{i,t}\times Treat_i$, $CF_{i,t}\times Post_{i,t}$, and $CF_{i,t}\times Treat_i\times Post_{i,t}$. Column (2) of Table 3 reports the results of equation (2). Consistent with our baseline findings, we continue to find a significantly positive effect of the closing auction on investment-q sensitivity, as indicated

[21] Untabulated results show no significant differences in matching variables between treatment and control groups.

by the coefficient of 0.220 ($p < 0.01$) on $Q_{i,t} \times Treat_i \times Post_{i,t}$. This result further mitigates concerns that our baseline results from equation (1) are driven by biases in staggered DiD estimations.

5. Tests of the feedback channel

In the preceding section, we document that investment-*q* sensitivity increases following the introduction of the NASDAQ closing auction. We interpret this finding as evidence that market feedback and managerial learning improve after the auction's implementation. Our proposed mechanism is that closing auction prices incorporate more private information about firm fundamentals, some of which is previously unknown to managers. This new information enables managers to update their expectations about investment opportunities, thereby improving investment efficiency. However, because learning is not directly observable, alternative explanations could also account for the observed increase in investment-*q* sensitivity.

In this section, we present a series of analyses consistent with the hypothesized feedback channel, thereby reinforcing the learning interpretation. First, we show that the closing auction enhances the information content of stock prices regarding future earnings. Second, we find that managers revise their beliefs more strongly in response to contemporaneous stock returns after the auction's implementation. Third, we provide evidence that the information driving the increase in investment-*q* sensitivity is likely new to managers. Finally, we document that the closing auction improves future operating performance, consistent with more efficient investment decisions.

5.1. Effects on future earnings response coefficient

The feedback channel posits that closing auction prices aggregate more private

information about firms' fundamentals from market participants. Consequently, we expect stock prices to better reflect future earnings following the implementation of the closing auction (Brogaard and Pan, 2022). To test this prediction, we examine the future earnings response coefficient (FERC), the extent to which current stock returns incorporate information about future earnings, following recent studies such as Israeli et al. (2017) and Lee and Watts (2021).

To estimate the effect of the closing auction on FERC, we follow prior studies (e.g., Israeli et al., 2017; Brogaard and Pan, 2022) and employ an augmented FERC model, as shown in equation (3). The model incorporates the staggered implementation of the closing auction (*Auction*) and its interactions with past, contemporaneous, and future earnings:

$$Return_{i,t} = \sum_{k=-1}^{1} (\beta_k \Delta Earnings_{i,t+k} \times Auction_{i,t-1} + \theta_k \Delta Earnings_{i,t+k} + \rho_k \Delta Earnings_{i,t+k} \times X_{i,t}) + \beta_2 Auction_{i,t-1} + \gamma X_{i,t} + \alpha_i + \eta_t + \varepsilon_{i,t} \quad (3)$$

where the subscripts i and t denote firms and quarters, respectively. $Return_{i,t}$ represents the buy-and-hold returns for quarter t, and $\Delta Earnings_{i,t+k}$ is the seasonal change in quarterly earnings per share excluding extraordinary items for quarter $t + k$ ($k = -1, 0, 1$), scaled by the beginning-of-quarter share price. $Auction_{i,t-1}$ equals one if the firm participated in the NASDAQ closing auction by the end of quarter $t-1$, and zero otherwise.

Following prior studies, we include controls ($X_{i,t}$) that may affect FERC: the natural logarithm of market capitalization ($MktCap_{i,t}$), institutional ownership ($InstHold_{i,t}$), quarterly asset growth ($Assetgr_{i,t}$), next quarter returns ($Return_{i,t+1}$), and an indicator for firms reporting negative earnings in the next quarter ($Loss_{i,t+1}$). We also include firm fixed effect (α_i) to control for time-invariant firm characteristics and quarter fixed effect (η_t) to capture time-varying shocks. The coefficient of interest, β_1 on $\Delta Earnings_{i,t+1} \times Auction_{i,t-1}$, measures the impact of

the closing auction on FERC.

Table 4 presents estimates from various specifications of equation (3). Column (1) includes only $\Delta Earnings_{i,t+1}$, $Auction_{i,t-1}$, their interaction, and control variables. Column (2) adds $\Delta Earnings_{i,t}$, $\Delta Earnings_{i,t-1}$, and their interactions with $Auction_{i,t-1}$. Column (3) further interacts control variables with $\Delta Earnings_{i,t+1}$. Column (4) presents the full specification. Across all columns, the coefficient on $\Delta Earnings_{i,t+1} \times Auction_{i,t-1}$ is significantly positive, indicating that prices become more informative about future earnings following the introduction of the closing auction. This evidence supports the notion that the auction mechanism enhances the aggregation of private information into stock prices.

5.2. Effects on managerial beliefs

A key premise of the feedback channel is that part of the additional private information aggregated through the closing auction is new to managers. Although such information is not directly observable, it can be reasonably inferred from changes in managerial beliefs, as reflected in revisions to management forecasts. Using management earnings forecasts, Zuo (2016) shows that when stock prices contain more private information, managers revise their annual earnings forecasts more strongly in response to contemporaneous stock returns. This finding provides compelling evidence of managerial learning by linking changes in managerial beliefs to market feedback. Building on this insight, we predict that if the closing auction aggregates more new information into stock prices, managers' forecast revisions will become more sensitive to contemporaneous stock returns following its implementation.

To test this prediction, we adapt Zuo's (2016) regression framework by including interaction terms that capture the effect of the closing auction:

$$ForecastRevision_{i,d_1,d_2} = \beta_1 Return_{i,d_1,d_2} \times Auction_{i,d_1} + \beta_2 Return_{i,d_1,d_2} + \beta_3 Auction_{i,d_1} + \gamma X_{i,d_1,d_2} + \alpha_j + \eta_{d_2} + \varepsilon_{i,d_1,d_2} \quad (4)$$

where $ForecastRevision_{i,d_1,d_2}$ represents the change in firm i's management earnings forecasts for fiscal year t between days d_1 and d_2, scaled by the closing price two days before d_1. $Return_{i,d_1,d_2}$ is the buy-and-hold return from one day after d_1 to one day before d_2. $Auction_{i,d_1}$ is an indicator variable equal to one if the closing auction had been implemented for firm i by day d_1, and zero otherwise. The coefficient of interest, β_1 on the interaction between $Return_{i,d_1,d_2}$ and $Auction_{i,d_1}$, captures the extent to which the closing auction affects the sensitivity of management forecast revisions to contemporaneous returns.

Following Zuo (2016), the control variables (X_{i,d_1,d_2}) include proxies for public information about future earnings, the consensus analyst forecast revision ($AnalystRevision_{i,d_1,d_2}$) and the cumulative quarterly management forecast error ($QuarterlyError_{i,d_1,d_2}$), and their interactions with $Auction_{i,d_1}$. $AnalystRevision_{i,d_1,d_2}$ is defined as the change in the prevailing consensus analyst forecasts between d_1 and d_2, and $QuarterlyError_{i,d_1,d_2}$ is the sum of management's quarterly forecast errors for any quarterly earnings announcement occurring between those days (detailed variable definitions are provided in Appendix B).

Additional controls include the natural logarithm of the forecast horizon ($Horizon_{i,d_2}$), the time gap between forecast dates (Gap_{i,d_1,d_2}), market capitalization ($MktCap_{i,d_1}$), analyst coverage ($Coverage_{i,d_1}$), and the market-to-book ratio (MTB_{i,d_1}). We include Fama-French 48-industry fixed effects (α_j) to capture industry-specific factors and year fixed effects (η_{d_2}), based on day d_2, to control for time trends. Non-indicator variables are winsorized at the 1st and 99th percentiles, and standard errors are clustered by industry and year (Zuo, 2016).

We estimate equation (4) using management annual earnings forecasts issued between 2002 and 2006 from the I/B/E/S Guidance database. Consistent with Zuo (2016), we include both point and range forecasts, using the reported value for point forecasts and the midpoint for range forecasts. Forecast revisions are identified when a firm issues multiple forecasts for the same annual earnings realization and the new forecast differs from the prior one. The final sample consists of 4,215 management forecast revisions for 543 NASDAQ firms with complete data for the variables in equation (4).

Panel A of Table 5 presents summary statistics. The mean values of $ForecastRevision_{i,d_1,d_2}$ and $Return_{i,d_1,d_2}$ are 0.164% and 0.031%, respectively. On average, the time between two management forecasts is 87 days (= $e^{4.466}$), and the average time from a forecast revision to the fiscal year-end is 105 days (= $e^{4.656}$). Panel B of Table 5 reports regression results for two specifications of equation (4). Column (1) includes a subset of control variables, while column (2) includes the full set. In both specifications, the coefficient on $Return_{i,d_1,d_2} \times Auction_{i,d_1}$ is significantly positive, indicating that management forecast revisions respond more strongly to contemporaneous stock returns after the implementation of the closing auction. This finding suggests that managers glean more *new* information from stock prices to update their beliefs following the implementation of the closing auction, consistent with the feedback channel.

5.3. Type of information new to managers

Another way to infer whether the information incorporated into prices is new to managers is to consider its nature. The feedback literature generally agrees that information pertaining to external factors beyond managerial control, such as macroeconomic conditions and

growth opportunities, is more likely to represent new knowledge to managers (Hutton, Lee, and Shu, 2012; Bai et al., 2016; Ye et al., 2023). Consistent with this view, we provide evidence that these types of external information drive the observed increase in investment-*q* sensitivity following the implementation of the closing auction.

5.3.1. Macroeconomic information

We begin by examining the role of macroeconomic information. Prior research suggests that macroeconomic information is more valuable for firms whose fundamentals are more exposed to macroeconomic factors (Hutton et al., 2012; Ye et al., 2023). If the closing auction enhances the incorporation of macroeconomic information into stock prices, we expect the increase in investment-*q* sensitivity to be stronger for firms with greater macroeconomic exposure.

To test this prediction, we construct two proxies for firm-level macroeconomic exposure, *Cyclicality* and *Energy*, following Hutton et al. (2012). *Cyclicality* (*Energy*) captures the extent to which a firm's earnings co-move with Gross Domestic Product (energy prices), with higher values indicating greater sensitivity to macroeconomic conditions.

We divide firms into two subgroups based on their average macroeconomic exposure during 2002Q1–2003Q4, the pre-auction period. We define an indicator variable, $High_i$ (Low_i), equal to one if a firm's pre-auction *Cyclicality* or *Energy* is above (below) the cross-sectional median, and zero otherwise. We then modify equation (1) by interacting $High_i$ and Low_i with all explanatory variables.

Columns (1) and (2) of Table 6 report results for *Cyclicality* and *Energy*, respectively. In both specifications, the coefficient on $High_i \times Q_{i,t} \times Auction_{i,t}$ is significantly positive, while

the coefficient on $Low_i \times Q_{i,t} \times Auction_{i,t}$ is insignificant. The difference between the two coefficients is statistically significant ($p < 0.01$). These results indicate that the increase in investment-q sensitivity following the introduction of the closing auction is concentrated among firms with higher macroeconomic exposure. This pattern is consistent with the feedback channel, suggesting that the closing auction facilitates greater aggregation of macroeconomic information into stock prices.

5.3.2. Growth opportunities

We next examine whether the closing auction aggregates more information about firms' growth opportunities into stock prices. The learning literature (e.g., Gao and Liang, 2013; Bai et al., 2016; Goldstein, 2023) suggests that managers are more likely to learn from market participants about growth opportunities than about assets in place. Accordingly, if the feedback channel operates through the aggregation of such information, the increase in investment-q sensitivity following the implementation of the closing auction should be more pronounced among growth firms.

To test this prediction, we divide firms into two subgroups based on their average market-to-book ratio during 2002Q1–2003Q4, the pre-auction period. We define an indicator variable, $High_i$ (Low_i), that equals one when firm i's pre-auction market-to-book ratio is above (below) the cross-sectional median, thereby identifying growth (value) firms. We then modify equation (1) by interacting $High_i$ and Low_i with all explanatory variables.

Column (3) of Table 6 reports the results. The coefficient on $High_i \times Q_{i,t} \times Auction_{i,t}$ is significantly positive, while the coefficient on $Low_i \times Q_{i,t} \times Auction_{i,t}$ is insignificant. The difference between the two coefficients is statistically significant ($p < 0.01$). These findings indicate

that the post-auction increase in investment-*q* sensitivity is concentrated among growth firms, consistent with the feedback channel's prediction that the closing auction enhances the aggregation of information about growth opportunities into stock prices.

5.4. Effects on firm operating performance

We next examine the implications of the feedback channel for firms' future operating performance. If managers glean more new information from stock prices following the implementation of the closing auction and use this information to guide investment decisions, we expect these investments to become more efficient, ultimately enhancing future operating performance.

To test this prediction, we adapt the regression model of Chen et al. (2007) to account for the staggered implementation of the closing auction ($Auction_{i,t}$):

$$Performance_{i,t+1} = \beta_1 Auction_{i,t} + \gamma X_{i,t} + \alpha_i + \eta_t + \varepsilon_{i,t+1} \quad (5)$$

where $Performance_{i,t+1}$ represents future operating performance, measured by return on assets (*ROA*) and operating income (*OI*). *ROA* is defined as income before extraordinary items scaled by lagged total assets, and *OI* is calculated as operating income after depreciation and amortization, also scaled by lagged total assets.

Table 7 reports the estimation results of equation (5) using *ROA* and *OI* as dependent variables in columns (1) and (2), respectively. The coefficient on $Auction_{i,t}$ is significantly positive in both specifications, indicating that the operating performance of NASDAQ-listed firms improves following the introduction of the closing auction. This finding supports the feedback channel hypothesis: by enhancing the aggregation of information into stock prices, the closing auction enables managers to make more informed and efficient investment decisions.

6. Additional Analyses

We perform a series of additional analyses to test the robustness of our main findings and to rule out alternative explanations for the observed increase in investment-*q* sensitivity following the closing auction.

We first examine whether our results are sensitive to alternative measures of Tobin's *q*. Because NASDAQ firms invest heavily in intangible assets, the conventional measure of Tobin's *q* may not fully capture their growth opportunities. To address this concern, we re-estimate equation (1) using total *q* and the corresponding investment measures developed by Peters and Taylor (2017). As reported in Panel A of Table OA1 in the Online Appendix, the increase in investment-*q* sensitivity remains significant when using these measures, confirming the robustness of our main results. We also construct an alternative version of Tobin's *q* based on the average closing price over the quarter rather than the closing price on the last trading day. Since investment decisions typically reflect price information accumulated over time, this alternative measure may better capture the relevant market signal. Consistent with this expectation, Panel B of Table OA1 shows that the increase in investment-*q* sensitivity continues to hold when using *q* based on average closing prices.

Next, we consider several alternative explanations for the observed increase in investment-*q* sensitivity to further reinforce the interpretation that it arises from enhanced information aggregation and managerial learning.

First, we examine whether the closing auction induces greater information production

by investors, which could, in turn, increase investment-*q* sensitivity.[22] Although we find weak evidence of increased analyst coverage, other proxies for information production, such as non-robotic EDGAR search volume (Lee and Watts, 2021) and analyst forecast accuracy (Gao and Huang, 2020), show no material change. Moreover, controlling for the effect of analyst coverage in the investment-*q* sensitivity regressions does not affect our main results (Panel A of Table OA2), suggesting that increased information production is not the primary mechanism behind our findings.

Second, we test whether reduced noise in stock prices following the introduction of the closing auction (e.g., Smith, 2006; Pagano et al., 2013) could explain the results. We control for price noise and its interaction with *Q* using two measures widely used in the microstructure literature: (1) the variance ratio (Lo and MacKinlay, 1988; Chordia et al., 2008), which captures deviations from a random walk, and (2) the correlation between closing auction returns and subsequent daily returns (e.g., Smith, 2006; Jegadeesh and Wu, 2022; Bogousslavsky and Muravyev, 2023), where more negative values indicate greater noise. As shown in Panel B of Table OA2, the effect of the closing auction on investment-*q* sensitivity remains significantly positive after controlling for these variables, indicating that reduced price noise does not explain our findings.

Finally, we examine whether changes in managerial private information could drive the

[22] The impact of the closing auction on information acquisition is theoretically ambiguous. On one hand, prior studies show that by consolidating liquidity, increasing competition in market making, and reducing bid–ask spreads, the closing auction lowers trading costs (Pagano et al., 2013; Bogousslavsky and Muravyev, 2023), which may stimulate informed trading and thus information acquisition (Chordia, Roll, and Subrahmanyam, 2008; Jayaraman and Milbourn, 2012; Fang, Noe, and Tice, 2009). On the other hand, Verrecchia (1982) demonstrates that traders' incentives to acquire information decrease as price informativeness increases. Because Section 5.1 shows that the closing auction enhances price informativeness through improved information aggregation, Verrecchia's (1982) model predicts that information acquisition may not rise following the auction's implementation.

results. If managers possessed less private information following the introduction of the closing auction, they might rely more on stock prices when making investment decisions. Using insider trading activity and absolute earnings announcement returns as proxies for managerial information (Chen et al., 2007), we find no evidence that changes in managerial information explain the increase in investment-*q* sensitivity (Panel C of Table OA2).

7. Conclusion

The real effects of secondary financial markets rest on two key premises: (1) market prices aggregate and reflect information from diverse participants, and (2) some of this information is not already known to managers. We exploit the staggered, mandatory implementation of NASDAQ's call auction for determining closing prices in 2004 as an exogenous shock to information aggregation. Our results show that the introduction of the closing auction increases investment-*q* sensitivity. We provide a portfolio of evidence consistent with a feedback channel through which the closing auction affects investment-*q* sensitivity: by aggregating more private information unknown to managers, the auction enhances managerial learning from stock prices. Supporting evidence includes higher FERC, stronger sensitivity of management earnings forecast revisions to contemporaneous stock returns, and improved future operating performance following the auction's implementation.

Overall, our findings underscore the importance of information aggregation, beyond information acquisition and production, in shaping the real effects of financial markets. In recent years, market structures have undergone substantial transformation through innovations such as high-frequency trading and dark pools, which alter the mechanisms by which information is incorporated into prices. Understanding how these developments influence real

corporate decisions through the information aggregation channel represents a fruitful avenue for future research.

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Appendix A. Phase-in schedule of NASDAQ closing auction

The table presents the number of unique NASDAQ-listed firms introduced to the closing auction on each specified date. Data source: https://www.nasdaqtrader.com/Trader.aspx?id=archiveheadlines&cat_id=2

Implementation date	# of stocks
April 12, 2004	10
April 19, 2004	10
April 26, 2004	10
May 3, 2004	50
May 10, 2004	50
May 17, 2004	106
May 19, 2004	26
May 24, 2004	201
June 2, 2004	196
June 14, 2004	796
June 15, 2004	200
June 24, 2004	1
June 25, 2004	1
July 14, 2004	2
July 15, 2004	1
July 16, 2004	4
July 19, 2004	1
July 20, 2004	3
July 21, 2004	3
July 22, 2004	3
July 23, 2004	2
September 30, 2004	1
October 1, 2004	1
November 30, 2004	34
December 6, 2004	854
December 13, 2004	855
Total	3,421

Appendix B. Variable definitions

Variables	Definition
Age	The natural logarithm of firm age, defined as the number of months since the firm's first trading date on monthly CRSP.
AnalystRevision	The difference between the consensus analyst forecasts on the two days of management earnings forecasts (for the same earnings realization), scaled by the closing price two days before the earlier management forecast date. It is expressed in percentage points.
Assetgr	The quarterly growth rate of total assets (ATQ), expressed in percentage points.
Auction	An indicator variable that equals one if a firm has been phased into the NASDAQ closing auction by the quarter end, and zero otherwise.
CAPX	Quarterly capital expenditures scaled by lagged total assets (ATQ), expressed in percentage points. Quarterly capital expenditures are calculated as Compustat CAPXY for the first fiscal quarter and as the change in CAPXY for the second, third, and fourth fiscal quarters.
CAPXRND	Quarterly total investment is defined as the sum of quarterly capital expenditures and quarterly research & development expenses (XRDQ) scaled by lagged total assets, expressed in percentage points. *XRDQ* is set to zero if it is missing.
CF	Quarterly cash flows (IBQ + DPQ) scaled by lagged total assets (ATQ), expressed in percentage points.
Coverage	The natural logarithm of the number of analysts following a firm immediately before the issuance of the earlier management earnings forecast.
Cyclicality	A firm's macroeconomic exposure to Gross Domestic Product (GDP) is defined as the *R*-square from the firm-level estimation of the following model over the 12 quarters prior to the implementation of NASDAQ closing auction: $EARN_{i,t}=\alpha_i+\beta_{i,GDP}GDP_t+\varepsilon_{i,t}$, where *EARN* represents the net income before extraordinary items (IBQ) and *GDP* represents the quarterly nominal GDP.[23]
Δ*Earnings*	The seasonal difference in earnings per share excluding extraordinary items (EPSPXQ), scaled by the share price at the end of the previous quarter (PRICE).
Energy	A firm's macroeconomic exposure to energy prices is defined as the *R*-square from the firm-level estimation of the following model over the 12 quarters prior to the implementation of NASDAQ closing auction: $EARN_{i,t}=\alpha_i+\beta_{i,Energy}Energy_t+\varepsilon_{i,t}$, where *EARN* represents the net income before extraordinary items (IBQ) and *Energy* represents the quarterly mean of the monthly fuel (energy) index.[24]
ForecastRevision	The difference between a management annual earnings forecast issued on a given day and the most recent management forecast for the same earnings realization, scaled by the closing price two days before the most recent forecast. It is expressed in percentage points.
Gap	The natural logarithm of the number of days between two management earnings forecasts.
Horizon	The natural logarithm of the number of days between the date of a management earnings forecast revision and the end of the forecasted period.
ILLIQ	The natural logarithm of Amihud's (2002) illiquidity measure. Amihud illiquidity is defined as $\frac{1}{T}\sum_{j=1}^{T}\frac{\lvert r_{i,j}\rvert}{DVOL_{i,j}}$, where T is the number of trading days in quarter t, $r_{i,j}$ represents the daily return, and $DVOL_{i,j}$ represents the dollar trading volume.

[23] The quarterly nominal GDP data are made available by the IMF Nominal GDP.
[24] The monthly fuel (energy) index is published by the International Monetary Fund and includes crude oil (petroleum), natural gas, coal price, and propane indices. This data can be accessed at IMF Monthly Fuel Index.

InstHHI	Institutional ownership concentration, measured using the Herfindahl-Hirschman Index.
InstHold	The proportion of outstanding shares held by institutional investors at quarter end, expressed in percentage points, estimated following Palacios, Moussawi, and Glushkov (2009).
*KZ*4	The four-variable Kaplan and Zingales (1997) score (excluding Q) as defined by Baker, Stein, and Wurgler (2003).
Leverage	Sum of long-term debt (DLTTQ) and current liabilities (DLCQ), scaled by total assets (ATQ).
Loss	An indicator that equals one if quarterly income before extraordinary items (IBQ) is negative, and zero otherwise.
MTB	The equity market-to-book ratio is defined as the market value of equity (PRICE×CSHOQ) divided by the book value of equity (CEQQ).
MktCap	The natural logarithm of the market value of common equity (PRICE× CSHOQ) at quarter-end.
OI	Operating income after depreciation and amortization (OIADPQ) scaled by lagged total assets (ATQ), expressed in percentage points.
Q	Tobin's Q, calculated as the market value of equity (PRICE×CSHOQ) plus the book value of assets (ATQ) minus the book value of equity (CEQQ), scaled by total assets (ATQ).
QuarterlyError	The sum of management's quarterly forecast errors associated with any quarterly earnings announcement occurring between the two forecast dates (excluding the day of the earlier management forecast but including the day of the later management forecast), scaled by the closing price two days before the issuance of the earlier management forecast. It is expressed in percentage points.
Return	For FERC analysis, *Return* is defined as quarterly buy-and-hold stock returns. For the analysis of management earnings forecast revisions, *Return* is defined as buy-and-hold stock returns from one day after the most recent management forecast to one day before the forecast revision (Zuo, 2016).
RND	Quarterly research & development expenses (XRDQ) scaled by lagged total assets, expressed in percentage points.
ROA	Net income (NIQ) scaled by lagged total assets (ATQ), expressed in percentage points.
Sale	Quarterly sale revenue (SALEQ) scaled by total assets (ATQ).
Salegr	The quarterly growth rate of sale revenue (SALEQ), expressed in percentage points.
Size	The natural logarithm of total assets (ATQ).
Tangibility	Quarterly net property, plant, and equipment (PPENTQ) scaled by total assets (ATQ), expressed in percentage points.

Figure 1
Mechanism of price formation in NASDAQ closing auction

This figure illustrates the price formation process in NASDAQ closing auction. Market-on-Close (MOC) buy and sell orders, being the most aggressively priced, receive the highest execution priority. Limit-on-Close (LOC) and Imbalance-Only (IO) buy and sell orders are ranked based on their limit prices, with buy orders sorted in descending order and sell orders in ascending order. The cumulative quantities of buy and sell orders create stepwise demand and supply functions, respectively. The closing price is determined at the intersection of the two stepwise functions, representing the market equilibrium.

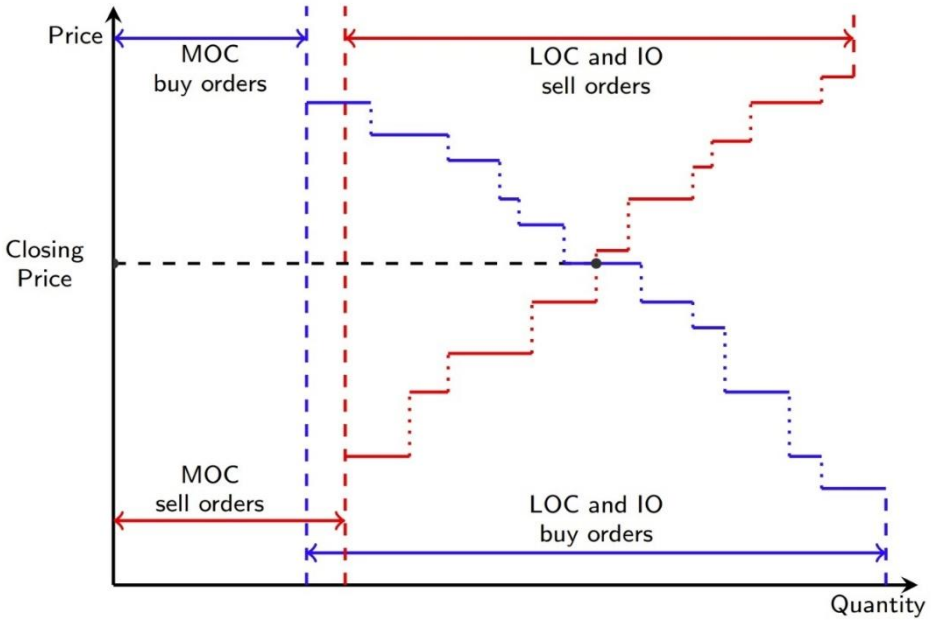


Figure 2
Dynamic Test of the change in investment-*q* sensitivity

This figure presents the results of an event-time analysis examining the impact of NASDAQ closing auction on investment-*q* sensitivity. Specifically, we re-estimate the regression model of investment-*q* sensitivity from Panel A of Table 2, replacing the dummy variable *Auction* with a series of indicators representing the quarters surrounding the implementation of the closing auction for each firm in the sample. In the figure, Quarter 0 represents the quarter in which the closing auction is implemented. Quarter $-k$ (k) refers to the kth quarter before (after) the auction's implementation. Quarter −4 (4) denotes four or more quarters before (after) the auction's implementation. The figure reports the coefficient estimates for the interactions between each event-quarter indicator and Tobin's *q*, along with their 90% confidence intervals. The quarter immediately preceding the implementation of the closing auction (Quarter −1) serves as the baseline for comparison. Panel A, B, and C present the results for total investment (*CAPXRND*), capital expenditures (*CAPX*), and R&D (*RND*), respectively.

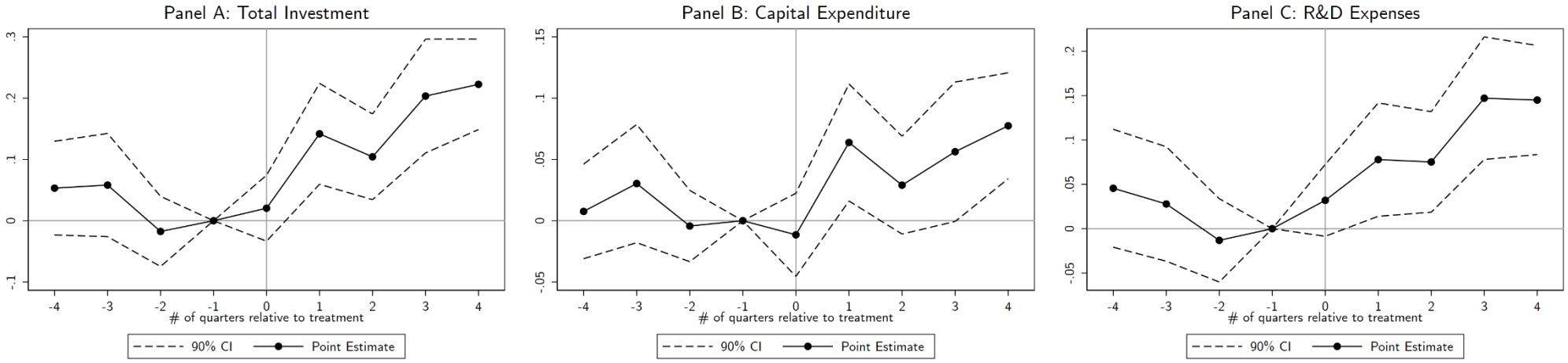


Table 1
Descriptive statistics

Variables	*N*	*Mean*	*SD*	*Q*1	*Median*	*Q*3
$CAPXRND_{i,t+1}$	28,521	3.290	3.321	0.951	2.390	4.384
$CAPX_{i,t+1}$	28,521	1.160	1.574	0.275	0.621	1.353
$RND_{i,t+1}$	28,521	2.130	3.177	0.000	0.896	3.080
$Auction_{i,t}$	28,521	0.451	0.498	0.000	0.000	1.000
$Q_{i,t}$	28,521	2.209	1.543	1.196	1.717	2.656
$CF_{i,t}$	28,521	0.234	5.965	−0.806	1.696	3.290
$Size_{i,t}$	28,521	5.173	1.348	4.177	5.081	6.058
$InstHold_{i,t}$	28,521	0.494	0.287	0.245	0.492	0.738
$ILLIQ_{i,t}$	28,521	−17.229	2.855	−19.304	−17.436	−15.258
$KZ4_{i,t}$	27,764	0.112	0.940	−0.576	−0.126	0.707
$Return_{i,t}$	28,470	0.050	0.303	−0.124	0.017	0.177
$\Delta Earnings_{i,t+1}$	27,965	0.011	0.093	−0.005	0.002	0.011
$MktCap_{i,t}$	28,521	5.439	1.535	4.372	5.406	6.409
$Assetgr_{i,t}$	28,521	3.160	20.945	−2.977	0.924	4.876
$Loss_{i,t+1}$	28,521	0.388	0.487	0.000	0.000	1.000
$ROA_{i,t+1}$	28,494	−0.959	6.080	−2.003	0.656	2.138
$OI_{i,t+1}$	28,371	0.066	5.514	−1.576	1.159	3.164
$Sale_{i,t}$	28,520	0.261	0.205	0.118	0.217	0.349
$InstHHI_{i,t}$	28,493	0.146	0.151	0.050	0.088	0.180

This table provides descriptive statistics for the variables in our main sample, with their definitions detailed in Appendix B.

Table 2
Effect of NASDAQ closing auction on investment-*q* sensitivity

Panel A: Baseline regression Dep. variable =	$CAPXRND_{i,t+1}$	$CAPX_{i,t+1}$	$RND_{i,t+1}$
	(1)	(2)	(3)
$\boldsymbol{Q_{i,t}\times Auction_{i,t}}$	**0.130*****	**0.038****	**0.092*****
	(0.03)	**(0.02)**	**(0.03)**
$Q_{i,t}$	1.659***	0.365***	1.294***
	(0.22)	(0.10)	(0.18)
$Auction_{i,t}$	−0.296***	−0.115*	−0.181**
	(0.09)	(0.06)	(0.07)
$CF_{i,t}$	−0.052***	0.001	−0.053***
	(0.01)	(0.00)	(0.01)
$CF_{i,t}\times Auction_{i,t}$	0.028***	0.010***	0.018**
	(0.01)	(0.00)	(0.01)
$Size_{i,t}$	−1.493***	−0.278***	−1.214***
	(0.12)	(0.05)	(0.11)
$Q_{i,t}\times Size_{i,t}$	0.026	0.055***	−0.029
	(0.02)	(0.01)	(0.02)
$InstHold_{i,t}$	0.276	0.451***	−0.175
	(0.30)	(0.17)	(0.23)
$Q_{i,t}\times InstHold_{i,t}$	0.078	−0.015	0.092
	(0.10)	(0.05)	(0.09)
$ILLIQ_{i,t}$	−0.182***	−0.139***	−0.043**
	(0.03)	(0.02)	(0.02)
$Q_{i,t}\times ILLIQ_{i,t}$	0.082***	0.028***	0.055***
	(0.01)	(0.01)	(0.01)
Firm FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Observations	28,517	28,517	28,517
Adjusted R^2	0.760	0.588	0.842

Table 2 (continued.)

Panel B: Robustness analysis with additional controls Dep. variable =	$CAPXRND_{i,t+1}$		
	(1)	(2)	(3)
$\boldsymbol{Q_{i,t}}\times\boldsymbol{Auction_{i,t}}$	**0.154*****	**0.129*****	**0.199*****
	(0.04)	**(0.03)**	**(0.04)**
$Q_{i,t}$	2.756***	1.598***	1.665***
	(0.31)	(0.23)	(0.22)
$Auction_{i,t}$	−0.313***	−0.281***	−0.404***
	(0.10)	(0.09)	(0.09)
$KZ4_{i,t}$		−0.252***	
		(0.09)	
$Q_{i,t}\times KZ4_{i,t}$		0.109***	
		(0.03)	
$Q_{i,t}\times MPD_Treat_i\times MPD_Post_t$			−0.336
			(0.25)
$Q_{i,t}\times MPD_Treat_i$			−0.205
			(0.24)
$Q_{i,t}\times MPD_Post_t$			−0.077**
			(0.04)
$MPD_Treat_i\times MPD_Post_t$			0.935*
			(0.56)
Q×Firm FE	Yes	No	No
Q×Group FE×Time trend	Yes	No	No
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Observations	28,517	27,758	28,517
Adjusted R^2	0.786	0.762	0.760

Panel A presents results of staggered DiD regressions examining the effect of NASDAQ closing auction on investment-q sensitivity. The dependent variable is total investment ($CAPXRND_{i,t+1}$), capital expenditures ($CAPX_{i,t+1}$), or R&D expenses ($RND_{i,t+1}$). The independent variable of interest is the interaction between Tobin's q ($Q_{i,t}$), measured at the end of quarter t, and an indicator variable that equals one if firm i had been phased into NASDAQ closing auction by that time ($Auction_{i,t}$). Panel B presents robustness tests of the effect of NASDAQ closing auction on investment-q sensitivity by further controlling for the interaction of $Q_{i,t}$ with firm fixed effects and group-specific time trends in column (1), the effect of financial constraints in column (2), and the effect of mandatory portfolio disclosure in column (3). Control variables (*Controls*) in Panel B include all other variables reported in Panel A. Definitions for all variables are provided in the Appendix B. Robust standard errors, clustered at the firm level, are reported in parentheses.

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 3

Effect of NASDAQ closing auction on investment-*q* sensitivity: Alternative specifications

Dep. variable =	$CAPXRND_{i,t+1}$	
	(1)	(2)
$\boldsymbol{Q_{i,t} \times Treat_i \times Post_{i,t}}$	**0.114****	**0.220*****
	(0.06)	**(0.07)**
$Q_{i,t}$	1.432***	0.428*
	(0.33)	(0.23)
$Q_{i,t} \times Treat_i$	0.181	−0.040
	(0.20)	(0.08)
$Q_{i,t} \times Post_{i,t}$	0.011	−0.005
	(0.04)	(0.03)
$Treat_i \times Post_{i,t}$	−0.170	−0.513***
	(0.15)	(0.13)
Controls	Yes	Yes
Cohort-Firm FE	Yes	No
Cohort-Quarter FE	Yes	No
Firm FE	No	Yes
Quarter FE	No	Yes
Observations	6,918	13,116
Adjusted R^2	0.837	0.664

This table presents results of the effect of NASDAQ closing auction on investment-*q* sensitivity using alternative regression specifications. In column (1), we estimate a stacked DiD regression using cohorts of treatment and control firms. Within each cohort, treatment firms ($Treat_i = 1$) include NASDAQ firms phased into the closing auction during April, May, or June 2024, while control firms ($Treat_i = 0$) include firms phased in during November and December 2024. The event window is [−1, +1], where quarter 0 refers to the quarter in which the treatment firms in the cohort implement the closing auction. $Post_{i,t}$ equals one for quarters 0 and 1, and zero for quarter −1. In column (2), we estimate a standard DiD regression on a matched sample, where NASDAQ-listed firm serve as the treatment group ($Treat_i = 1$) and NYSE-listed firms serve as the control group ($Treat_i = 0$). Each NASDAQ-listed firm is matched to a NYSE-listed firm in the same Fama-French 48 industry based on market capitalization, Tobin's *q*, and index fund ownership at the end of the last quarter before the NASDAQ firm entered the closing auction. For the treatment group, $Post_{i,t}$ equals one if firm *i* had entered the closing auction by quarter *t*, and zero otherwise. For the control group, $Post_{i,t}$ equals one if the matched NASDAQ firm had entered the closing auction by quarter *t*, and zero otherwise. Control variables (*Controls*) include all other variables reported in Panel A of Table 2. Robust standard errors, clustered at the firm level, are reported in parentheses.

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 4
Effect of NASDAQ closing auction on future earnings response coefficients

Dep. variable =	$Return_{i,t}$			
	(1)	(2)	(3)	(4)
$\Delta Earnings_{i,t+1} \times Auction_{i,t-1}$	**0.207*****	**0.192*****	**0.157****	**0.162****
	(0.07)	**(0.07)**	**(0.07)**	**(0.07)**
$\Delta Earnings_{i,t+1}$	0.107***	0.020	0.042	−0.020
	(0.03)	(0.03)	(0.08)	(0.09)
$Auction_{i,t-1}$	−0.024**	−0.026***	−0.026***	−0.027***
	(0.01)	(0.01)	(0.01)	(0.01)
$\Delta Earnings_{i,t} \times Auction_{i,t-1}$		0.157**	0.155**	0.207***
		(0.07)	(0.07)	(0.07)
$\Delta Earnings_{i,t}$		0.300***	0.307***	0.391***
		(0.03)	(0.03)	(0.10)
$\Delta Earnings_{i,t-1} \times Auction_{i,t-1}$		0.108	0.105	0.113
		(0.08)	(0.08)	(0.08)
$\Delta Earnings_{i,t-1}$		0.182***	0.185***	0.347***
		(0.03)	(0.03)	(0.08)
$MktCap_{i,t}$	0.198***	0.195***	0.195***	0.193***
	(0.01)	(0.01)	(0.01)	(0.01)
$InstHold_{i,t}$	−0.621***	−0.588***	−0.588***	−0.571***
	(0.03)	(0.03)	(0.03)	(0.03)
$Assetgr_{i,t}$	0.000***	0.000***	0.000***	0.000***
	(0.00)	(0.00)	(0.00)	(0.00)
$Loss_{i,t+1}$	−0.038***	−0.037***	−0.034***	−0.030***
	(0.01)	(0.01)	(0.01)	(0.01)
$Return_{i,t+1}$	0.021**	0.009	0.011	0.002
	(0.01)	(0.01)	(0.01)	(0.01)
$\Delta Earnings_{i,t+1} \times Controls$	No	No	Yes	Yes
$\Delta Earnings_{i,t} \times Controls$	No	No	No	Yes
$\Delta Earnings_{i,t-1} \times Controls$	No	No	No	Yes
Firm FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes
Observations	27,888	27,473	27,473	27,473
Adjusted R^2	0.261	0.276	0.276	0.282

This table presents the results of regressions examining the effect of NASDAQ closing auction on future earnings response coefficient (FERC). The dependent variable is the current quarter's buy-and-hold return ($Return_{i,t}$). The independent variable of interest is the interaction between the subsequent quarter's seasonal difference in earnings ($\Delta Earnings_{i,t+1}$) and an indicator variable that equals one if firm i had been phased into the NASDAQ closing auction by the end of quarter t–1 ($Auction_{i,t-1}$). Definitions of all variables are provided in Appendix B. Robust standard errors, clustered at the firm level, are reported in parentheses.

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 5
Effect of NASDAQ closing auction on managerial beliefs

Panel A: Descriptive statistics

Variables	*N*	*Mean*	*SD*	*Q*1	*Median*	*Q*3
$ForecastRevision_{i,d_1,d_2}$	4,215	0.164	2.638	0.000	0.008	0.357
$Auction_{i,d_1}$	4,215	0.524	0.500	0.000	1.000	1.000
$Return_{i,d_1,d_2}$	4,215	0.031	0.219	−0.086	0.018	0.132
$AnalystRevision_{i,d_1,d_2}$	4,215	0.129	2.424	−0.049	0.041	0.383
$QuarterlyError_{i,d_1,d_2}$	4,215	0.003	0.022	0.000	0.000	0.000
$MktCap_{i,d_1}$	4,215	6.357	1.272	5.575	6.214	6.960
MTB_{i,d_1}	4,215	1.066	0.634	0.636	1.058	1.464
Gap_{i,d_1,d_2}	4,215	4.466	0.787	4.111	4.511	4.718
$Coverage_{i,d_1}$	4,215	1.927	0.797	1.386	1.946	2.485
$Horizon_{i,d_2}$	4,215	4.656	0.900	3.761	4.868	5.394

Panel B: Effect of NASDAQ closing auction on management forecast revisions

Dep. variable =	$ForecastRevision_{i,d_1,d_2}$	
	(1)	(2)
$Return_{i,d_1,d_2} \times Auction_{i,d_1}$	**0.447****	**0.426***
	(0.16)	**(0.19)**
$Return_{i,d_1,d_2}$	0.613**	0.636*
	(0.26)	(0.29)
$Auction_{i,d_1}$	0.037	0.067
	(0.11)	(0.11)
$AnalystRevision_{i,d_1,d_2} \times Auction_{i,d_1}$	−0.108	−0.109
	(0.09)	(0.09)
$AnalystRevision_{i,d_1,d_2}$	0.774***	0.772***
	(0.07)	(0.07)
$QuarterlyError_{i,d_1,d_2} \times Auction_{i,d_1}$	−0.861	−1.151
	(1.02)	(0.96)
$QuarterlyError_{i,d_1,d_2}$	0.708	0.834
	(0.99)	(1.02)
$MktCap_{i,d_1}$		−0.028
		(0.05)
MTB_{i,d_1}		0.184**
		(0.07)
Gap_{i,d_1,d_2}		0.021
		(0.04)
$Coverage_{i,d_1}$		−0.014
		(0.06)
$Horizon_{i,d_2}$		0.032
		(0.03)
Industry FE	Yes	Yes
Year FE	Yes	Yes
Observations	4,214	4,214
Adjusted R^2	0.460	0.461

Panel A presents descriptive statistics for the sample of management earnings forecast revisions. Panel B presents regression results examining the effect of NASDAQ closing auction on the sensitivity of managerial earnings forecast revisions ($ForecastRevision_{i,d_1,d_2}$) to contemporaneous stock returns ($Return_{i,d_1,d_2}$). The independent variable of interest is the interaction between stock returns ($Return_{i,d_1,d_2}$) and an indicator variable that equals one if firm *i* had been phased into the NASDAQ closing auction by the time of the earlier management earnings forecast

($Auction_{i,d_1}$). Definitions of all variables are provided in Appendix B. Robust standard errors, clustered at the industry and revision announcement year levels, are reported in parentheses.
***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 6
Cross-sectional variations in the increase of investment-*q* sensitivity

Dep. variable =	$CAPXRND_{i,t+1}$		
Partition variable =	*Cyclicality*	*Energy*	*MTB*
	(1)	(2)	(3)
$\mathbf{High_i \times Q_{i,t} \times Auction_{i,t}}$	**0.192*****	**0.193*****	**0.128*****
	(0.04)	**(0.04)**	**(0.04)**
$\mathbf{Low_i \times Q_{i,t} \times Auction_{i,t}}$	**0.037**	**0.071**	**0.051**
	(0.04)	**(0.04)**	**(0.14)**
p-value for difference	*< 0.01*	*< 0.01*	*< 0.01*
$High_i \times Q_{i,t}$	1.856***	1.732***	1.234***
	(0.30)	(0.31)	(0.24)
$Low_i \times Q_{i,t}$	1.509***	1.600***	1.902***
	(0.30)	(0.29)	(0.41)
$High_i \times Auction_{i,t}$	−0.391***	−0.466***	−0.148
	(0.11)	(0.11)	(0.13)
$Low_i \times Auction_{i,t}$	−0.160	−0.150	−0.246
	(0.11)	(0.11)	(0.22)
$High_i \times Controls$	Yes	Yes	Yes
$Low_i \times Controls$	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Observations	28,517	28,517	28,517
Adjusted R^2	0.760	0.760	0.763

This table examines cross-sectional variation in the change in investment-*q* sensitivity following the implementation of the NASDAQ closing auction. The sample is partitioned based on firms' macroeconomic exposure, measured by the comovement of earnings with nominal Gross Domestic Product (*Cyclicality*) or with the energy price index (*Energy*), and growth opportunities, measured by the market-to-book ratio (*MTB*). Firms are classified into two groups, $High_i$ and Low_i, where $High_i$ (Low_i) equals one if a firm's pre-auction average of the partitioning variable (measured over 2002Q1–2003Q4) is above (below) the cross-sectional median, and zero otherwise. The dummy variables $High_i$ and Low_i are interacted with Tobin's *q* ($Q_{i,t}$) and an indicator variable for the implementation of the closing auction ($Auction_{i,t}$). Definitions of all variables are provided in Appendix B. The coefficient of interest is the difference between the coefficients on $High_i \times Q_{i,t} \times Auction_{i,t}$ and that on $Low_i \times Q_{i,t} \times Auction_{i,t}$. The reported *p*-value for this difference is based on 1,000 bootstrap iterations. Control variables (*Controls*) include all other variables reported in Panel A of Table 2. Robust standard errors clustered at the firm level are reported in parentheses.

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 7
Effect of NASDAQ closing auction on firm operating performance

Dep. variable =	$ROA_{i,t+1}$	$OI_{i,t+1}$
	(1)	(2)
***Auction*$_{i,t}$**	**0.522*****	**0.328*****
	(0.17)	**(0.12)**
$Sale_{i,t}$	8.717***	5.928***
	(0.92)	(0.85)
$InstHHI_{i,t}$	−1.193**	−1.455***
	(0.57)	(0.47)
$Q_{i,t}$	0.222***	0.178***
	(0.07)	(0.06)
$KZ4_{i,t}$	−0.769***	−0.669***
	(0.12)	(0.10)
Firm FE	Yes	Yes
Quarter FE	Yes	Yes
Observations	27,703	27,582
Adjusted R^2	0.576	0.752

This table presents regression results examining the effect of NASDAQ closing auction on firm operating performance. The dependent variable is return on assets ($ROA_{i,t+1}$) in column (1) and operating income ($OI_{i,t+1}$) in column (2). The independent variable of interest is an indicator that equals one if firm *i* had been phased into the NASDAQ closing auction by the end of quarter *t* ($Auction_{i,t}$). Definitions of all variables are provided in Appendix B. Robust standard errors, clustered at the firm level, are reported in parentheses.
***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.